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Bearing inner race with sleeve and integrated puller

Abstract

A bearing assembly for a turbomachine includes a bearing inner race, a bearing outer race located radially outwardly from the bearing inner race, and one or more bearing elements located radially between the bearing inner race and the bearing outer race. A bearing sleeve is positioned radially inboard of the bearing inner race and is configured for engagement with a rotating component. The bearing sleeve includes a sleeve body located radially between the bearing inner race and the rotating component, and a puller feature extending from the sleeve body at a first axial end of the bearing sleeve. The puller feature is configured for application of a removal force thereto to remove the bearing from an installed position in the turbomachine.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2690346	12/1953	Miller	384/538	F16C 35/073
3953142	12/1975	Price	384/538	F16D 1/096
4456425	12/1983	McCarty	74/416	F16H 1/14
5373636	12/1993	Martinie	29/256	B25B 27/062
7356926	12/2007	Bridges et al.	N/A	N/A
9409266	12/2015	Buldtmann	N/A	F01D 25/285
2005/0259905	12/2004	Hansen	384/538	F16C 35/073
2009/0297083	12/2008	Raberin	384/537	F16C 35/073
2014/0130352	12/2013	Buldtmann	29/888.3	F01D 25/285
2015/0226265	12/2014	Bourgeois	29/259	B66C 1/62
2017/0226894	12/2016	Ganiger et al.	N/A	N/A
2018/0038378	12/2017	DiBenedetto	N/A	N/A
2021/0332775	12/2020	Yuen et al.	N/A	N/A
2022/0127973	12/2021	Lagarde	N/A	F01D 25/16
2022/0290617	12/2021	Marlin	N/A	F02C 7/36

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202013008469	12/2014	DE	N/A
2010169150	12/2009	JP	F16C 25/083

OTHER PUBLICATIONS

Gorla, Rama and Aijaz Khan. Turbomachinery Design and Theory. New Yor, NY, Marcel Dekker, 2003. Chapter 1, section 1.2. (Year: 2003). cited by examiner
Extended European Search Report for EP Application No. 24183513.1, dated Dec. 11, 2024, pp. 1-6. cited by applicant

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Background/Summary

BACKGROUND

(1) Exemplary embodiments pertain to the art of turbomachinery such as gas turbine engines. In particular, the present disclosure relates to bearing assemblies of gas turbine engines.

(2) Turbomachines, such as gas turbine engines typically include one or more bearing assemblies to radially and/or axially support the rotating components of the turbomachine. The bearing assemblies typically include radially inner and outer rings or races, rolling elements such as rollers or balls disposed between the inner and outer races, and a cage to keep the rolling elements separated and help guide their motion. The inner race is typically press fit onto a rotating component, such as a shaft, while the outer race is typically a close push fit to a rotationally stationary component, such as a wall or flange of a bearing housing.

(3) To achieve installation of the bearing, in particular installation of the inner race to the shaft a force is applied to a first race surface in the installation direction to press fit the inner race to the shaft. When it is desired to disassemble the bearing, a removal force is applied to a second race surface, opposite the first race surface, at an opposite end of the inner race to pull the inner race from the shaft. To access the second race surfaces, a specialized tool is often needed to reach the surface and allow for applying the removal force. Because of the various designs of bearing assemblies and the compartments in which they are located, these removal tools are different for each bearing assembly. Further, in some turbomachine configurations or bearing assembly locations, there is insufficient access to reach the second race surface even with a specialized tool.

BRIEF DESCRIPTION

(4) In one embodiment, a bearing assembly for a turbomachine includes a bearing inner race, a bearing outer race located radially outwardly from the bearing inner race, and one or more bearing elements located radially between the bearing inner race and the bearing outer race. A bearing sleeve is positioned radially inboard of the bearing inner race and is configured for engagement with a rotating component. The bearing sleeve includes a sleeve body located radially between the bearing inner race and the rotating component, and a puller feature extending from the sleeve body at a first axial end of the bearing sleeve. The puller feature is configured for application of a removal force thereto to remove the bearing from an installed position in the turbomachine.

(5) Additionally or alternatively, in this or other embodiments the first axial end of the bearing sleeve is further configured for application of an installation force thereto to install the bearing into the installed position.

(6) Additionally or alternatively, in this or other embodiments the puller feature is a sleeve flange extending radially outwardly from the sleeve body.

(7) Additionally or alternatively, in this or other embodiments a locating protrusion extends from the sleeve body and is engageable with the bearing inner race to axially locate the bearing sleeve relative to the bearing inner race.

(8) Additionally or alternatively, in this or other embodiments the axial location of the bearing sleeve relative to the bearing inner race results in an axial gap between the bearing inner race and the puller feature.

(9) Additionally or alternatively, in this or other embodiments the bearing sleeve is press fit between the rotating component and the bearing inner race.

(10) Additionally or alternatively, in this or other embodiments the bearing sleeve is a full unitary ring.

(11) Additionally or alternatively, in this or other embodiments the puller feature is formed integral to the sleeve body.

(12) Additionally or alternatively, in this or other embodiments one or more of the bearing inner

race and the bearing outer race are full unitary rings.

(13) Additionally or alternatively, in this or other embodiments the one or more bearing elements are one or more spherical balls or cylindrical rollers.

(14) In another embodiment, a turbomachine includes one or more rotating shafts, one or more turbines operably connected to the one or more rotating shafts and one or more bearing assemblies, each bearing assembly supportive of a shaft of the one or more rotating shafts. A bearing assembly of the one or more bearing assemblies includes a bearing inner race, a bearing outer race positioned radially outwardly from the bearing inner race, and one or more bearing elements located radially between the bearing inner race and the bearing outer race. A bearing sleeve is located radially inboard of the bearing inner race and configured for engagement with the shaft. The bearing sleeve includes a sleeve body positioned radially between the bearing inner race and the shaft, and a puller feature extending from the sleeve body at a first axial end of the bearing sleeve. The puller feature is configured for application of a removal force thereto to remove the bearing from an installed position in the turbomachine.

(15) Additionally or alternatively, in this or other embodiments the first axial end of the bearing sleeve is further configured for application of an installation force thereto to install the bearing into the installed position.

(16) Additionally or alternatively, in this or other embodiments the puller feature is a sleeve flange extending radially outwardly from the sleeve body.

(17) Additionally or alternatively, in this or other embodiments a locating protrusion extends from the sleeve body and is engageable with the bearing inner race to axially locate the bearing sleeve relative to the bearing inner race.

(18) Additionally or alternatively, in this or other embodiments the axial location of the bearing sleeve relative to the bearing inner race results in an axial gap between the bearing inner race and the puller feature.

(19) Additionally or alternatively, in this or other embodiments the bearing sleeve is press fit between the shaft and the bearing inner race.

(20) Additionally or alternatively, in this or other embodiments the bearing sleeve is a full unitary ring.

(21) Additionally or alternatively, in this or other embodiments the puller feature is formed integral to the sleeve body.

(22) Additionally or alternatively, in this or other embodiments one or more of the bearing inner race and the bearing outer race are full unitary rings.

(23) Additionally or alternatively, in this or other embodiments the one or more bearing elements are one or more spherical balls or cylindrical rollers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a partial cross-sectional view of a gas turbine engine; and

(3) FIG. 2 is a partial cross-sectional view of an embodiment of a bearing of a gas turbine engine.

DETAILED DESCRIPTION

(4) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(5) FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 is disclosed herein as a two-spool turbofan that generally incorporates a fan section 22, a compressor section 24, a combustor section 26 and a turbine section 28. Alternative engines might include other

systems or features. The fan section **22** drives air along a bypass flow path B in a bypass duct, while the compressor section **24** drives air along a core flow path C for compression and communication into the combustor section **26** then expansion through the turbine section **28**. Although depicted as a two-spool turbopfan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbopfans as the teachings may be applied to other types of turbine engines including three-spool architectures.

(6) The exemplary engine **20** generally includes a low speed spool **30** and a high speed spool **32** mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **36** via several bearing systems **38**. It should be understood that various bearing systems **38** at various locations may alternatively or additionally be provided, and the location of bearing systems **38** may be varied as appropriate to the application.

(7) The low speed spool **30** generally includes an inner shaft **40** that interconnects a fan **42**, a low pressure compressor **44** and a low pressure turbine **46**. The inner shaft **40** is connected to the fan **42** through a speed change mechanism, which in exemplary gas turbine engine **20** is illustrated as a geared architecture **48** to drive the fan **42** at a lower speed than the low speed spool **30**. The high speed spool **32** includes an outer shaft **50** that interconnects a high pressure compressor **52** and high pressure turbine **54**. A combustor **56** is arranged in exemplary gas turbine **20** between the high pressure compressor **52** and the high pressure turbine **54**. An engine static structure **36** is arranged generally between the high pressure turbine **54** and the low pressure turbine **46**. The engine static structure **36** further supports bearing systems **38** in the turbine section **28**. The inner shaft **40** and the outer shaft **50** are concentric and rotate via bearing systems **38** about the engine central longitudinal axis A which is collinear with their longitudinal axes.

(8) The core airflow is compressed by the low pressure compressor **44** then the high pressure compressor **52**, mixed and burned with fuel in the combustor **56**, then expanded over the high pressure turbine **54** and low pressure turbine **46**. The turbines **46**, **54** rotationally drive the respective low speed spool **30** and high speed spool **32** in response to the expansion. It will be appreciated that each of the positions of the fan section **22**, compressor section **24**, combustor section **26**, turbine section **28**, and fan drive gear system **48** may be varied. For example, gear system **48** may be located aft of combustor section **26** or even aft of turbine section **28**, and fan section **22** may be positioned forward or aft of the location of gear system **48**.

(9) The engine **20** in one example is a high-bypass geared aircraft engine. In a further example, the engine **20** bypass ratio is greater than about six (6), with an example embodiment being greater than about ten (10), the geared architecture **48** is an epicyclic gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3 and the low pressure turbine **46** has a pressure ratio that is greater than about five. In one disclosed embodiment, the engine **20** bypass ratio is greater than about ten (10:1), the fan diameter is significantly larger than that of the low pressure compressor **44**, and the low pressure turbine **46** has a pressure ratio that is greater than about five 5:1. Low pressure turbine **46** pressure ratio is pressure measured prior to inlet of low pressure turbine **46** as related to the pressure at the outlet of the low pressure turbine **46** prior to an exhaust nozzle. The geared architecture **48** may be an epicycle gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3:1. It should be understood, however, that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present disclosure is applicable to other gas turbine engines including direct drive turbopfans.

(10) A significant amount of thrust is provided by the bypass flow B due to the high bypass ratio. The fan section **22** of the engine **20** is designed for a particular flight condition—typically cruise at about 0.8 Mach and about 35,000 feet (10,688 meters). The flight condition of 0.8 Mach and 35,000 ft (10,688 meters), with the engine at its best fuel consumption—also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFC’)”—is the industry standard parameter of lbf of

fuel being burned divided by lbf of thrust the engine produces at that minimum point. “Low fan pressure ratio” is the pressure ratio across the fan blade alone, without a Fan Exit Guide Vane (“FEGV”) system. The low fan pressure ratio as disclosed herein according to one non-limiting embodiment is less than about 1.45. “Low corrected fan tip speed” is the actual fan tip speed in ft/sec divided by an industry standard temperature correction of $[(T_{\text{fan}} \text{ } ^\circ\text{R}) / (518.7^\circ \text{ R})]^{0.5}$. The “Low corrected fan tip speed” as disclosed herein according to one non-limiting embodiment is less than about 1150 ft/second (350.5 m/sec).

(11) Referring now to FIG. 2, illustrated is an exemplary embodiment of a bearing **38**. The exemplary bearing **38** may be located at, for example, the inner shaft **40** or the outer shaft **50**. In the following description, the shaft is an inner shaft **40**, but one skilled in the art will readily appreciate that this is merely exemplary, and that the shaft may be an outer shaft **50** or any other shaft or rotating component. The bearing **38** is located in a housing **60** located at a radially outer extent of the bearing **38**, with components of the bearing **38** being disposed inside, or radially inboard of, a housing inner wall **62** of the housing **60**. A bearing outer race **64** is located inside the housing **60** at the housing inner wall **62**. The bearing outer race **64**, in some embodiments, is a full ring extending a full rotation about the engine central longitudinal axis A. In some embodiments, the bearing outer race **64** has a close radial fit to the housing inner wall **62**. A bearing inner race **66** is located at the inner shaft **40** and in some embodiments is a full ring extending a full rotation about the engine central longitudinal axis A. One or more bearing elements **68** are disposed radially between the bearing outer race **64** and the bearing inner race **66** and are arrayed circumferentially about the engine central longitudinal axis A. The bearing elements **68** may be, for example, spherical balls or may be cylindrical rollers. These configurations of bearing elements **68** are, however, merely exemplary and one skilled in the art will readily appreciate that other bearing elements shapes may be utilized. The bearing elements **68** contact the bearing outer race **64** and the bearing inner race **66** to support the shaft **40** relative to the housing **60**. In some embodiments, the bearing elements **68** are retained in a cage **70**, to maintain a selected circumferential spacing between the bearing elements **68** and to also guide the motion of the bearing elements **68**.

(12) A bearing sleeve **72** is disposed radially between a radially inner surface **74** of the bearing inner race **66** and a shaft outer surface **76** of the shaft **40**, and in some embodiments has a press fit between the bearing inner race **66** and the shaft **40**. The bearing sleeve **72**, in some embodiments, is a full ring extending circumferentially about the engine central longitudinal axis A between the bearing inner race **66** and the shaft **40**.

(13) The bearing **38** is configured for installation between the housing **60** and the shaft **40** in a first axial direction, referred to herein as installation direction **78** and is configured for removal from its installed position in a second axial direction, or removal direction **80** opposite the installation direction **78**. The bearing sleeve **72** is configured to aid in removal of the bearing **38** in the removal direction **80**. To this end, the bearing sleeve **72** includes a sleeve body **82** disposed between the shaft outer surface **76** and the bearing inner race **66**. The sleeve body **82** is configured to have a press fit to the shaft outer surface **76** and to the bearing inner race **66**. The bearing sleeve **72** further includes an integral puller feature, such as a sleeve flange **84** disposed at a body end **86** of the sleeve body **82**, and extending radially outwardly from the sleeve body **82**, relative to the engine central longitudinal axis A. While in the embodiment illustrated the puller feature is the sleeve flange **84**, in other embodiments the puller feature may have another configuration, such as a hook, tab or slot feature to which a removal tool, may be connected. In some embodiments, the sleeve body **82** includes a locating protrusion **90** to axially locate the bearing sleeve **72** on the bearing inner race **66**, which in some embodiments leaves an axial gap **92** between the bearing inner race **66** and the sleeve flange **84** allowing the removal tool to be engaged with the sleeve flange **84** between the sleeve flange **84** and the bearing inner race **66**.

(14) To install the bearing **38** to the shaft **40**, an installation or pushing force is applied to the body end **86** or alternatively to the sleeve flange **84** to move the bearing **38** in the installation direction

78 until the bearing 38 is in the installed position as shown in FIG. 2. To disassemble the bearing 38 from the shaft 40, a removal or pulling force is applied to the sleeve flange 84 via, for example, the removal tool to pull the bearing 38 in the removal direction 80. The puller feature, such as the sleeve flange 84, allows for both the removal force and the installation force to be applied at the same axial end of the bearing 38. This allows for ease of removal of the bearing 38 in configurations where the opposing axial end of the bearing is not accessible for applying the removal force. Further, use of the bearing sleeve 72, allows for use of the same bearing on shafts 40 having different diameters, requiring changes only to the bearing sleeve 72 to fit the bearing 38 to the shaft 40. Additionally, use of the bearing sleeve 72 allows for simpler designs of components around the bearing 38 since accessibility to the opposing axial end of the bearing 38 for removal is not a concern.

(15) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5%, or 2% of a given value.

(16) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(17) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A bearing and shaft assembly of a turbomachine, comprising: a bearing inner race; a bearing outer race disposed radially outwardly from the bearing inner race; one or more bearing elements located radially between the bearing inner race and the bearing outer race; and a bearing sleeve disposed radially inboard of the bearing inner race and configured for engagement with a shaft supporting a rotor of the turbomachine, the bearing sleeve including: a sleeve body disposed radially between the bearing inner race and the rotor shaft; a puller feature extending from the sleeve body at a first axial end of the bearing sleeve, the puller feature configured for application of a removal force thereto to remove the bearing from an installed position in the turbomachine; and a locating protrusion extending from the sleeve body and engageable with the bearing inner race to axially locate the bearing sleeve relative to the bearing inner race, the locating protrusion positioned axially between the puller feature and the bearing inner race; wherein the locating protrusion axially abuts the bearing inner race so that a channel is defined by the puller feature, the bearing inner race, and the locating protrusion.

2. The bearing assembly of claim 1, wherein the first axial end of the bearing sleeve is further configured for application of an installation force thereto to install the bearing into the installed position.

3. The bearing assembly of claim 1, wherein the puller feature is a sleeve flange extending radially outwardly from the sleeve body.

4. The bearing assembly of claim 1, wherein the bearing sleeve is press fit between the rotor and the bearing inner race.
 5. The bearing assembly of claim 1, wherein the bearing sleeve is a full unitary ring.
 6. The bearing assembly of claim 1, wherein the puller feature is formed integral to the sleeve body.
 7. The bearing assembly of claim 1, wherein one or more of the bearing inner race and the bearing outer race are full unitary rings.
 8. The bearing assembly of claim 1, wherein the one or more bearing elements are one or more spherical balls or cylindrical rollers.
 9. A gas turbine engine, comprising: one or more rotating shafts; one or more turbines operably connected to the one or more rotating shafts; one or more bearing assemblies, each bearing assembly supportive of a corresponding shaft of the one or more rotating shafts, a bearing assembly of the one or more bearing assemblies, including: a bearing inner race; a bearing outer race disposed radially outwardly from the bearing inner race; one or more bearing elements located radially between the bearing inner race and the bearing outer race; and a bearing sleeve disposed radially inboard of the bearing inner race and engaged with the corresponding shaft, the bearing sleeve including: a sleeve body disposed radially between the bearing inner race and the corresponding shaft; a puller feature extending from the sleeve body at a first axial end of the bearing sleeve, the puller feature configured for application of a removal force thereto to remove the bearing from an installed position in the turbomachine; and a locating protrusion extending from the sleeve body and engageable with the bearing inner race to axially locate the bearing sleeve relative to the bearing inner race, the locating protrusion positioned axially between the puller feature and the bearing inner race; wherein the locating protrusion axially abuts the bearing inner race so that a channel is defined by the puller feature, the bearing inner race, and the locating protrusion.
 10. The gas turbine engine of claim 9, wherein the first axial end of the bearing sleeve is further configured for application of an installation force thereto to install the bearing into the installed position.
 11. The gas turbine engine of claim 9, wherein the puller feature is a sleeve flange extending radially outwardly from the sleeve body.
 12. The gas turbine engine of claim 9, wherein the bearing sleeve is press fit between the shaft and the bearing inner race.
 13. The gas turbine engine of claim 9, wherein the bearing sleeve is a full unitary ring.
 14. The gas turbine engine of claim 9, wherein the puller feature is formed integral to the sleeve body.
 15. The gas turbine engine of claim 9, wherein one or more of the bearing inner race and the bearing outer race are full unitary rings.
 16. The gas turbine engine of claim 9, wherein the one or more bearing elements are one or more spherical balls or cylindrical rollers.
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